Supplementary Material: Benchmark Definitions, Execution Protocol, and Uncertainty

The Grid-GDP functions are adapted from the Quantum Hamiltonian Descent collection; each instance is defined by its implemented expression and domain rather than its conventional name (SECQUOIA, 2026a; Wu Group, n.d.).

S1 Coordinate domains and normalization

The coordinate mapping and objective normalization follow Section 3.4 of the main paper. Both coordinate endpoints are included, and L and U are the minimum and maximum over the 961 lattice points. All trigonometric arguments are in radians. Full-precision normalization constants and target coordinates are archived in `grid_inventory.json` (Section S8).

Table S1: Grid-GDP benchmark domains and target-set sizes. Each problem uses a 31×31 lattice with 961 configurations and 62 Boolean indicators. n_{target} counts lattice points with normalized objective $F \leq 10^{-5}$.

Function	x domain	y domain	n_{target}
Ackley	$[-5, 5]$	$[-5, 5]$	1
Alpine 1	$[0, 10]$	$[0, 10]$	1
Alpine 2	$[0, 1000]$	$[0, 1000]$	1
Bohachevsky 2	$[-100, 100]$	$[-100, 100]$	1
Three-hump camel	$[-5, 5]$	$[-5, 5]$	1
Csendes (squared variant)	$[-1, 1]$	$[-1, 1]$	36
Deflected corrugated spring	$[0, 10]$	$[0, 10]$	1
Drop-Wave	$[-5.12, 5.12]$	$[-5.12, 5.12]$	1
Easom	$[-10, 10]$	$[-10, 10]$	1
Griewank	$[-600, 600]$	$[-600, 600]$	1
Hölder table	$[-10, 10]$	$[-10, 10]$	4
Hosaki	$[0, 5]$	$[0, 6]$	1
Lévy	$[-10, 10]$	$[-10, 10]$	2
Lévy 13	$[-10, 10]$	$[-10, 10]$	4
Michalewicz	$[0, \pi]$	$[0, \pi]$	1
Rastrigin	$[-5.12, 5.12]$	$[-5.12, 5.12]$	1
Rosenbrock	$[-2, 2]$	$[-1, 3]$	2
Shubert (variant)	$[0, 10]$	$[0, 10]$	1
Styblinski–Tang	$[-5, 5]$	$[-5, 5]$	1
Sum of squares	$[-10, 10]$	$[-10, 10]$	1
Xin-She Yang 3 (variant)	$[-20, 20]$	$[-20, 20]$	870

S2 Grid-GDP objective functions

The implemented functions below use the domains in Table S1 (SECQUOIA, 2026a).

1. Ackley

$$f(x,y) = -20\exp\left(-0.2\sqrt{\frac{x^2+y^2}{2}}\right) - \exp\left(\frac{\cos(2\pi x)+\cos(2\pi y)}{2}\right) + 20 + e. \tag{S1}$$

2. Alpine 1

$$f(x,y) = |x\sin x + 0.1x| + |y\sin y + 0.1y|. \tag{S2}$$

3. Alpine 2

$$f(x,y) = -\sqrt{xy}\,\sin x \sin y. \tag{S3}$$

4. Bohachevsky 2

$$f(x,y) = x^2 + 2y^2 - 0.3\cos(3\pi x)\cos(4\pi y) + 0.3. \tag{S4}$$

5. Three-hump camel

$$f(x,y) = 2x^2 - 1.05x^4 + \frac{x^6}{6} + xy + y^2. \tag{S5}$$

6. Csendes (squared variant)

$$f(x,y) = c(x) + c(y), \qquad c(t) = \begin{cases} t^6\left[2+\sin(1/t)\right]^2, & t \neq 0, \\ 0, & t = 0. \end{cases} \tag{S6}$$

The GDP uses branch-fixed auxiliaries $r_t = 1/t$ for $t \neq 0$ and $r_t = 0$ otherwise, avoiding division by zero without an epsilon shift.

7. Deflected corrugated spring

With $r = \sqrt{(x-5)^2 + (y-5)^2}$,

$$f(x,y) = 0.1r^2 - \cos(5r). \tag{S7}$$

8. Drop-Wave

$$f(x,y) = -\frac{1+\cos\left(12\sqrt{x^2+y^2}\right)}{0.5(x^2+y^2)+2}. \tag{S8}$$

9. Easom

$$f(x,y) = -\cos x \cos y\, \exp\left(-[(x-\pi)^2 + (y-\pi)^2]\right). \tag{S9}$$

10. Griewank

$$f(x,y) = \frac{x^2+y^2}{4000} - \cos x \cos\left(\frac{y}{\sqrt{2}}\right) + 1. \tag{S10}$$

11. Hölder table

$$f(x,y) = -\left|\sin x \cos y\, \exp\left(\left|1 - \frac{\sqrt{x^2+y^2}}{\pi}\right|\right)\right|. \tag{S11}$$

12. Hosaki

$$f(x,y) = \left(1 - 8x + 7x^2 - \frac{7}{3}x^3 + \frac{1}{4}x^4\right) y^2 \exp(-y). \tag{S12}$$

13. Lévy

Define $w_x = 1 + (x-1)/4$ and $w_y = 1 + (y-1)/4$. Then

$$\begin{aligned} f(x,y) = {} & \sin^2(\pi w_x) \\ & + (w_x - 1)^2\big[1 + 10\sin^2(\pi w_x + 1)\big] \\ & + (w_y - 1)^2\big[1 + \sin^2(2\pi w_y)\big]. \end{aligned} \tag{S13}$$

14. Lévy 13

$$\begin{aligned} f(x,y) = {} & \sin^2(3\pi x) + (x-1)^2\big[1 + \sin^2(3\pi y)\big] \\ & + (y-1)^2\big[1 + \sin^2(2\pi y)\big]. \end{aligned} \tag{S14}$$

15. Michalewicz

$$f(x,y) = -\sin x\, \sin^{20}\left(\frac{x^2}{\pi}\right) - \sin y\, \sin^{20}\left(\frac{2y^2}{\pi}\right). \tag{S15}$$

16. Rastrigin

$$f(x,y) = x^2 - 10\cos(2\pi x) + y^2 - 10\cos(2\pi y) + 20. \tag{S16}$$

17. Rosenbrock

$$f(x,y) = (1-x)^2 + 100(y - x^2)^2. \tag{S17}$$

18. Shubert (variant)

$$\begin{aligned} f(x,y) = {} & \big[\cos(2x+1) + 2\cos(3x+2) + 3\cos(4x+3)\big] \\ & \times \big[\cos(2y+1) + \cos(y+2)\big]. \end{aligned} \tag{S18}$$

19. Styblinski–Tang

$$f(x,y) = \frac{1}{78}\,\frac{1}{2}\left(x^4 - 16x^2 + 5x + y^4 - 16y^2 + 5y\right). \tag{S19}$$

20. Sum of squares

$$f(x,y) = x^2 + 2y^2. \tag{S20}$$

21. Xin-She Yang 3 (variant)

$$f(x,y) = \exp\left(-\left[\left(\frac{x}{15}\right)^6 + y^6\right]\right) + 2\exp\big(-(x^2+y^2)\big)\cos^2 x \cos^2 y. \tag{S21}$$

S3 Executable protocol and implementation settings

Python and NumPy use the job seed, and `RBFOpt` also receives `rand_seed`. `BAM` exposes no separate native random-seed option: its process wrapper seeds Python/NumPy, while the Grid seed identifies the common initial design. `FMQA-SA` uses three NumPy `SeedSequence(seed)` streams for FM initialization, annealing, and uniform unseen-point fallback; each training or annealing invocation draws a seed from its stream. `PyTorch` uses deterministic CPU operations and one thread.

Grid `D-SDA` enumerates L_∞ neighbors lexicographically from $(-1, 0, 1)$, excluding the zero direction and out-of-bounds points; ties select the first point.

The `RBFOpt` options are `rbf=auto`, `global_search_method=genetic`, `do_init_strategy=False`, `ga_base_population_size=400`, and `ga_num_generations=20`.

`BAM` receives `x0=None` and `max_evals`=B-3; its local models use `ALAMO`/`BARON`, whose internal versions are not reported by the package.

The `FMQA-SA` time limit is checked between operations and through training/sampling callbacks. Each FM uses identity activation, zero bias, and normal initial weights with mean zero and standard deviation 0.01. Training uses Adam (Kingma and Ba, 2015), 1,000 full-batch epochs, mean squared error, and learning rate 0.01. Coordinate ξ_j uses $\lfloor \log_2 u_j \rfloor + 1$ unsigned bits, where $u_j > 0$ is its upper bound; QUBO coefficients are divided by the largest absolute entry of the quadratic matrix. D-Wave's `SimulatedAnnealingSampler` in `dwave-samplers` 1.6.0 uses 100 reads, 1,000 sweeps, and a geometric inverse-temperature schedule (D-Wave Systems, n.d.). All process workers and the Grid `FMQA-SA` worker have a 2,100 s watchdog. The Grid `RBFOpt`/`BAM` trace watchdogs are disabled (`max_run_seconds=0`); archived Grid `D-SDA` uses budget, target, or local-search termination without an additional method-clock limit.

S3.1 Versions and independently timed process evaluations

The recorded Python version is 3.11.16; other versions are given in the main paper.

The conventional GDP process models retain their archived initializations, including constructor defaults for `CSTR30`; the updated DFO evaluation table is identified separately in Section S8.

The acceptance rule in `gdp_models/evaluation.py` requires all of the following: a finite objective; termination in {`optimal`, `locallyOptimal`, `globallyOptimal`}; solver status other than `error` or `aborted`; interface status `SOLVESTAT`=1 and `MODELSTAT`$\in \{1, 2\}$; and a finite maximum primal violation no greater than 10^{-5}. Time-limit, iteration-limit, evaluation-limit, and error outcomes are not accepted, even if a finite feasible candidate is available.

The primal check uses unscaled violations of original constraints and initialized-variable bounds, including constraint rows deactivated during preprocessing. Nonfinite objectives, constraint bodies, or initialized-variable values preclude acceptance; unused uninitialized variables are skipped. Preprocessing rejection or worker failure supplies no accepted objective.

The `CSTR` initializer `cstr_box_ipopt.py:initialize_feasible` runs after preprocessing and checks the mass-balance/zero-recycle start at 10^{-8} without changing constraints or bounds. All process evaluations use cold starts; NLP solutions are not transferred between configurations. Objectives, acceptance status, and NLP point costs are linked to the same recorded attempt by model, `CSTR` size, and coordinate, without successful-retry selection or cross-size time substitution.

S3.2 Observed proposal and initialization behavior

Table S2 pools `FMQA-SA` proposals across budgets and seeds, excluding the shared initialization and including terminal callbacks. Fallback supplies 52.89% of process proposals; the uniform comparison evaluates the complete procedure, without isolating the contributions of learning, annealing, and fallback.

Table S2: Recorded `FMQA-SA` uniform fallback by benchmark family.

Family	Runs	Proposals	Fallbacks	Rate
Grid	1,050	51,450	9,522	18.51%
`CSTR`	60	1,350	369	27.33%
`Column`	10	100	86	86.00%
`Dynamic`	60	1,460	1,063	72.81%
`Small Batch`	10	70	58	82.86%
Process total	140	2,980	1,576	52.89%

S4 Process model scale, units, and discrete decisions

Table S3 counts the original GDP before reformulation or presolve, including all active disjunct branches and fixed continuous variables. `Dynamic` is discretized with 30 finite elements per stage and three Radau points. Binary counts refer to the indicator variables already constructed by the model, before Boolean-to-binary conversion. All nine models have zero additional nonbinary integer variables.

Table S3: Original process-model sizes. n_c: continuous variables; n_b: binary variables; m_a: algebraic constraints; m_l: logical constraints; n_d: disjunctions.

Model	n_c	n_b	m_a	m_l	n_d
`CSTR5`	56	20	100	12	10
`CSTR10`	106	40	195	22	20
`CSTR15`	156	60	290	32	30
`CSTR20`	206	80	385	42	40
`CSTR25`	256	100	480	52	50
`CSTR30`	306	120	575	62	60
`Column`	446	30	637	16	15
`Small Batch`	19	18	34	3	9
`Dynamic`	1647	27	3276	31	9

The `Column` objective is $10^3(Q_c + Q_b) + 10^3 n_{\text{active}}$, the source model's weighted cost proxy rather than a newly estimated annualized cost. The `Dynamic` objective is $-\int_0^9 x(t)^2\,dt$.

S5 Lomustine simulation-based benchmark

The `Lomustine` configurations and responses come from the corrected-model tables of SECQUOIA (2026b), generated with PharmaPy at commit 3194efd. For each superstructure option, the source ran `PyNomadBBO` 4.6.0 from a stored initial point on the normalized operating inputs with 60 simulations and stored the best objective, the constraint values, the equipment sizes, and the wall time of the inner optimization. The inner search accepts a point when the solubility path constraint holds. The heptane mass-fraction specification of the source is not enforced, because the fixed three-hour vaporizer operation cannot reach it within the pressure bounds. Table S7 lists the 36 two-reactor configurations; the four single-reactor semibatch options of the source do not fit the three external variables and are excluded. The configurations with a batch crystallizer have the lowest objective for every reactor pair, and the choice of the first reactor changes the objective by up to a factor of 2.9. The vendored table, case module, run records, and analysis are listed in Section S8.

Option order and objective. The external variables index categorical alternatives, so we repeated the 210 runs with the five other orders of Table S4, using the same seeds, initial designs in index space, and budgets. Table S5 reports the hits in the source order and their range over the six orders. `NOMAD` 4.6.0 ended with a segmentation fault after identical evaluations on both attempts in 12 runs over all campaigns; as for the Grid runs of Section S12, these runs are scored on the evaluations completed before the crash. Table S6

repeats the source-order runs with the bare-module capital cost of each flowsheet at its optimized operation as the objective, whose reference is the all-batch flowsheet at \$82,766. This cost is evaluated at the operating point that maximizes production and crystal size, and it is not itself optimized over the operating inputs.

Table S4: `Lomustine`: orders of the reactor and crystallizer alternatives used as external-variable indices; crystallizers are listed as batch or by the number of MSMPR stages.

Order	Reactors	Crystallizers
Source	batch, CSTR, PFR	batch, 1, 2, 3
Reversed	PFR, CSTR, batch	3, 2, 1, batch
Back-mixing	batch, PFR, CSTR	batch, 3, 2, 1
Random 1	PFR, batch, CSTR	1, 2, batch, 3
Random 2	batch, PFR, CSTR	batch, 3, 1, 2
Random 3	CSTR, PFR, batch	2, 1, batch, 3

Table S5: `Lomustine`: runs out of ten that reach the reference configuration in the source order and, in brackets, the range over the six orders of Table S4.

Method	$B = 10$	$B = 15$	$B = 20$
`D-SDA`	1 [1, 6]	4 [3, 8]	4 [3, 8]
`NOMAD`	6 [5, 6]	6 [5, 9]	6 [5, 9]
`RBFOpt`	8 [2, 8]	9 [7, 10]	10 [9, 10]
`MISO`	7 [4, 7]	10 [5, 10]	10 [6, 10]
`BAM`	0 [0, 2]	4 [0, 10]	6 [0, 10]
`FMQA-SA`	2 [2, 7]	3 [3, 10]	7 [7, 10]
Uniform	2 [0, 4]	4 [2, 4]	5 [4, 7]

Table S6: `Lomustine` with the capital-cost objective: runs out of ten that reach the all-batch reference configuration, in the source order. Bold counts mark the largest number of hits for each budget, including ties.

Method	$B = 10$	$B = 15$	$B = 20$
`D-SDA`	3	6	9
`NOMAD`	9	9	9
`RBFOpt`	7	8	**10**
`MISO`	8	**10**	**10**
`BAM`	**10**	**10**	**10**
`FMQA-SA`	2	3	5
Uniform	4	7	8

Table S7: `Lomustine`: objective of every configuration and, in parentheses, the wall time of its simulation-based optimization. Rows give the first (R01) and second (R02) reactor, and columns the crystallizer. The bold entry is the reference configuration.

R01	R02	batch	1 MSMPR	2 MSMPR	3 MSMPR
batch	batch	-170.68 (91 s)	-131.70 (310 s)	-146.86 (538 s)	-148.92 (835 s)
batch	CSTR	-230.52 (146 s)	-175.25 (348 s)	-175.27 (650 s)	-166.13 (970 s)
batch	PFR	-268.46 (159 s)	-209.36 (348 s)	-213.29 (651 s)	-202.26 (975 s)
CSTR	batch	-483.85 (152 s)	-360.16 (331 s)	-365.91 (656 s)	-345.28 (936 s)
CSTR	CSTR	-331.90 (159 s)	-244.39 (367 s)	-245.34 (660 s)	-239.50 (1019 s)
CSTR	PFR	-429.81 (206 s)	-315.22 (408 s)	-321.46 (717 s)	-304.62 (1017 s)
PFR	batch	**-487.84** (208 s)	-362.45 (347 s)	-361.55 (655 s)	-348.06 (886 s)
PFR	CSTR	-351.71 (183 s)	-249.82 (388 s)	-259.47 (659 s)	-244.65 (983 s)
PFR	PFR	-433.41 (211 s)	-318.34 (393 s)	-324.18 (629 s)	-306.21 (907 s)

S6 Catalytic distillation column

Corrected model. We checked every equation block, parameter table, and bound of the published model against standard forms and literature values, and the `Catalytic Column` benchmark uses the model with the errors fixed. The corrections concern the vapor–liquid equilibrium and the equilibrium constant of the reaction, the mixture liquid density (Rackett equation with the Spencer–Danner mixing rule) and surface tension (Macleod–Sugden), the tray hydraulics, and the objective, which counted the annualization factor twice and credited the impurities in the bottoms at the ETBE price. The pure-component data of isobutene, 1-butene, and ethanol are fitted to open reference equations of state, and the ETBE data, for which no reference equation exists, agree with open measurements of its vapor pressure and density. The bottoms must contain at least 95 mol% ETBE, the fuel-grade specification, the catalytic stages stay at or below 120 °C, the maximum operating temperature of the Amberlyst 15 catalyst, and the prices are 2010 market prices. The repository lists each finding with its effect on the published optimum; with the 95 mol% specification, the published optimal configuration is infeasible, because its column is too short.

Evaluation campaign. The 210 runs made 12,015 evaluations: 6,921 returned accepted objectives, 2,789 were logical rejections without an NLP solve, and 2,305 returned unaccepted NLP responses. Eight `D-SDA` runs stopped before their budgets because no neighbor improved, and all other runs used their full budgets. The runs requested 3,499 distinct configurations, of which 2,652 were accepted, and solving them took 7.7 h. The runs are charged 20.2 h of completion time in total, but the campaign took 2.8 h on three cores because a configuration requested by several runs was solved once. Each run evaluated distinct configurations, so the reuse comes from different runs requesting the same configuration.

Search box. The GDP admits 100,947 configurations, and the gap box $g_i \leq 6$ of the search methods keeps 54,159 of them, including every design within 5% of the best of a pilot of 202 configurations. The reference has $g_1 = f_1 - 2 = 6$, on the bound of the box, so a better design could lie just outside it. We therefore solved, with the initialization and acceptance rule of the campaign, all 747 configurations outside the box within two stages of the reference in every stage position. Of these, 255 are accepted and 492 are locally infeasible, and none is better than the reference: the best, $(9, 9, 13, 17, 17, 21)$ at 24,930.96 \$/yr, lies 0.14% above it. This check ran on a workstation with `GAMS` 54.2.2 and `CONOPT` 4.39.1, the `CONOPT` version of the campaign, and every solve finished within 97 s, below the campaign's 300 s limit.

MINLP forms. The published MINLP form makes the stage decisions binary variables, multiplies switched equations by these variables or by the stage-existence indicator, and keeps the property and kinetic equations on every stage. The disjunctive form behind the Big-M and Hull reformulations has five disjunctions for each internal stage, one per decision, whose terms collect the equations that the GDP model switches on or off (Table S8). The property and feed-enthalpy equations hold on every stage, as in the published form, but the kinetics stay in the catalyst disjunction; with the kinetics on every stage, the bounds of the reaction rate restricted the temperature of a stage without catalyst, and with the binary variables of $(3, 4, 5, 11, 11, 14)$ fixed, the Big-M reformulation stopped at 32,075 \$/yr, a point from which the GDP form continues to 30,822 \$/yr. Each nonlinear disjunct equation $h(\mathbf{x}) = 0$ or $h(\mathbf{x}) \leq 0$ is replaced by the definition $s\,r = h(\mathbf{x})$, which holds on every stage, and by the linear disjunct constraint $r = 0$ or $r \leq 0$; a linear disjunct equation is divided by s. The scale s is 1.01 times the largest absolute value of h over the variable bounds, computed by interval arithmetic after feasibility-based bound tightening with the constraints that hold for every configuration. Every scaled disjunct constraint therefore lies in $[-1, 1]$ at every feasible point, so $M = 1$ is a valid Big-M value for all of them, and the bound tightening adds 101 implied variable bounds that are written into the model. The scales range from 0.10 to 1.3×10^{18}; the largest belongs to the rate equation, whose products of activities, adsorption constant, and rate span this range over the variable bounds. The interval bounds of the flooding and weeping correlations are finite only if the liquid is denser than the vapor, which the source bounds do not imply, so the disjunctive form bounds the liquid and vapor mass densities by $\rho_L \geq 200$ and $5 \leq \rho_V \leq 150$ kg/m^3; every validated solution has $\rho_L \geq 529$ and $\rho_V \leq 34$ kg/m^3. The `GAMS` convex hull reformulation accepts only inequalities, so each disjunct equality is written as two inequalities, and Big-M of the pair equals Big-M of the equality. Table S9 compares the forms with the binary variables fixed.

Table S8: Disjunctions of each internal stage k in the disjunctive form of the `Catalytic Column`.

Indicator	Indicator true	Indicator false
stage k exists	vapor–liquid equilibrium, tray hydraulics, pressure drop, flooding and weeping limits, tray height	vapor composition and flow pass through, no pressure drop, no tray height
catalyst on stage k	kinetics, catalyst hydraulics and flooding, reaction terms, 120 °C limit	no reaction, no catalyst pressure drop
ethanol feed at k	feed flow	no feed flow
butenes feed at k	feed flow	no feed flow
boil-up return at k	boil-up flow	no boil-up flow

Table S9: Objective (\$/yr) of each form with the binary variables fixed, solved by `CONOPT` from the initialization of the published model. "Infeasible" marks configurations that every form reports as locally infeasible.

$(f_1, c_1, c_2, c_3, f_2, b)$	GDP	Published MINLP	Big-M	Hull
$(8, 8, 12, 17, 17, 21)$	24,895.55	24,895.55	24,895.55	24,895.55
$(7, 8, 11, 16, 16, 20)$	25,271.33	25,271.33	25,271.33	25,271.33
$(3, 4, 5, 11, 11, 14)$	30,822.02	30,822.02	30,822.02	30,822.02
$(4, 4, 9, 10, 14, 17)$	28,373.98	28,373.98	28,373.98	28,373.98
$(3, 6, 12, 14, 19, 21)$	31,116.60	32,320.55	31,116.60	31,116.60
$(2, 3, 9, 10, 20, 21)$	35,745.43	35,194.65	35,745.43	35,745.43
$(4, 4, 6, 8, 8, 10)$		infeasible		
$(3, 4, 5, 6, 8, 9)$		infeasible		
$(13, 13, 15, 16, 17, 19)$		infeasible		

MINLP solvers. Table S10 lists the results of the MINLP solvers. `SBB` stops on the Big-M and Hull reformulations when the NLP relaxations of its nodes exceed the limit on function evaluations outside their domains. Table S11 lists the global optimality certificates that `SCIP` and `BARON` return when they start from

a feasible design of the published form; the GAMS Examiner confirms that both starting designs satisfy every constraint and bound within 10^{-6}.

Table S10: MINLP solvers on the Catalytic Column from the initial configuration $(3, 4, 5, 6, 8, 9)$ of the published model, one core, 3,600 s limit. The reference is 24,895.55 \$/yr, and times are wall-clock times of the GAMS job.

Form	Solver	Outcome	Time (s)
Published	DICOPT	no feasible design; every NLP subproblem infeasible, iteration limit	686
	SBB	28,717.22 at $(4, 4, 5, 6, 6, 9)$	10
	BARON	declared infeasible	255
	SCIP	declared infeasible	9
Big-M	DICOPT	no feasible design; NLP subproblem failed	902
	SBB	no feasible design; domain violation limit	38
	BARON	declared infeasible	214
	SCIP	no feasible design; lower bound -5.7×10^7	3,600+
Hull	DICOPT	no feasible design; NLP subproblem failed	211
	SBB	no feasible design; domain violation limit	130
	BARON	declared infeasible	362
	SCIP	no feasible design; lower bound -5.7×10^7	3,600+

Table S11: Global optimality certificates for the published MINLP form started from two feasible designs.

Starting design	SCIP	BARON
$(7, 8, 11, 16, 16, 20)$	optimal at 25,271.33 (8 s)	optimal at 25,271.33 (172 s)
$(8, 8, 12, 17, 17, 21)$	optimal at 24,895.55 (10 s)	optimal at 24,895.55 (270 s)

S7 Completion costs

Table S12 reports completion costs for all 14 process conditions, together with the cost of evaluating every configuration of each DFO domain once. Table S13 reports the Grid-GDP costs at $B = 100$. MISO reaches the most Grid targets, but its median search time of 32.70 s is about four times that of RBFOpt, whereas D-SDA and NOMAD stop after medians of 37 and 31.5 evaluations and have the lowest median completion times, 18.38 and 18.50 s (NOMAD over its 189 timed runs).

Per-function Grid results at $B = 100$ and complete seed-level outcomes are available in the analysis files linked in Section S8.

Table S12: Median reconstructed full-run completion costs for all ten paired seeds per condition, including target misses, and the cost of full enumeration, in seconds.

Case	B	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform	Enumeration
Column	13	35.44	31.93	54.60	31.98	27.50	41.82	21.69	137.24
CSTR5	8	10.67	10.71	10.72	11.33	10.67	13.92	9.44	30.27
CSTR10	15	17.61	21.25	21.79	21.87	18.54	27.07	17.75	122.06
CSTR15	22	25.68	33.92	37.59	39.79	27.67	42.82	27.84	298.42
CSTR20	29	32.31	38.17	45.03	56.03	39.37	59.10	40.12	553.38
CSTR25	36	45.49	56.56	62.29	78.56	50.47	74.70	51.17	899.39
CSTR30	43	47.17	71.23	101.96	107.33	115.41	101.19	65.20	1719.00
Small Batch	10	4.23	4.97	5.12	5.62	4.23	9.09	3.49	10.18
Dynamic	12	12.22	12.36	12.29	11.80	9.95	14.47	7.80	47.25
Dynamic	15	15.32	15.48	16.10	15.44	11.53	18.39	8.71	47.25
Dynamic	20	20.02	20.85	20.54	42.42	14.07	25.23	12.27	47.25
Dynamic	30	20.02	22.00	34.19	106.24	21.91	36.24	19.38	47.25
Dynamic	37	20.02	22.00	42.42	111.61	24.99	46.05	24.38	47.25
Dynamic	50	20.02	21.99	65.14	195.36	33.98	61.31	33.12	47.25

Table S13: Grid-GDP results at $B = 100$, with 210 runs per method (21 functions and ten seeds). Target hits reach the normalized reference zero within 10^{-5}. Times are Q50 (median) [Q25, Q75] in seconds across all runs, including target misses, using linearly interpolated quartiles. Summed configuration-evaluation cost includes the separately measured cost of every evaluation in the full run, including repeated configurations. Search time excludes lookup processing; adding it to the summed configuration-evaluation cost gives estimated full-run completion time. Component quartiles need not sum to completion quartiles. [a]NOMAD times cover the 189 runs with recorded times; the 21 runs that crashed on every attempt have none, but their target hits are counted (Section S12).

Method	Target hits	Full-run completion (s)	Summed configuration-evaluation cost (s)	Search time (s)
D-SDA	101/210	18.38 [12.42, 37.06]	18.38 [12.41, 37.05]	0.0024 [0.0023, 0.0026]
NOMAD[a]	126/210	18.50 [13.46, 35.00]	15.20 [11.09, 30.82]	2.04 [1.79, 2.28]
RBFOpt	171/210	70.81 [54.83, 93.09]	60.84 [44.31, 82.15]	7.58 [4.12, 14.82]
MISO	181/210	94.82 [79.22, 116.05]	63.21 [44.03, 82.37]	32.70 [25.19, 39.38]
BAM	147/210	64.42 [48.74, 81.67]	63.97 [48.34, 81.22]	0.43 [0.37, 0.51]
FMQA-SA	129/210	124.13 [105.57, 139.26]	62.34 [43.98, 77.52]	62.04 [61.58, 62.53]
Uniform	45/210	60.51 [43.33, 78.60]	60.51 [43.33, 78.59]	0.0026 [0.0026, 0.0027]
BAM-prefix control	124/210	61.81 [40.96, 78.33]	61.80 [40.95, 78.33]	0.0026 [0.0026, 0.0027]

S8 Data sources and reproducibility

The publication snapshot contains the source code, experimental data, analysis results, and reproduction instructions for every experiment in this article. The `CSTR` updates are the penalty of 10^6 for unaccepted responses, analyzed in `results/cstr_penalty_1e6_analysis_20261022`, and the `CSTR30` evaluation table rebuilt with the mass-balance/zero-recycle initialization, in `results/cstr30_publication_20261025`; the `Column` runs on its 64-configuration logical domain with $B = 13$ are in `results/column_logical_b13_analysis_20261025`. The updated GDPopt analysis and execution records contain the 63 Grid and 27 process runs configured with `SCIP` for MIP and `IPOPT` for NLP subproblems. The three presolve-disabled LOA diagnostic runs are excluded. The same records contain the paired comparison with the earlier configuration that used `SCIP` for both MIP and NLP subproblems, in which GLOA returned the same `CSTR` objectives within a relative difference of 3×10^{-5}. Seed spread, tolerance sensitivity, search overhead, and `GDPopt` enumeration counts are generated by `review_comparison/revision_diagnostics.py` from the retained run records and execution logs. The `D-SDA` rerun, including the process restart variants, is `experiments/dsda_no_revisit/run.py`, which replays the sealed lookup package of the adopted `D-SDA` rerun. The `NOMAD` and `MISO` runs were executed at source revision c4d7db6; the returned records contain the original archive, its SHA-256 checksum, and hash-checked analysis tables of every run. All `D-SDA` runs, including the restart variants, were rerun at source revision c186408, where no configuration is evaluated twice; with re-evaluation switched back on, the same code reproduces all 1,190 published `D-SDA` runs exactly, and the returned records hold every run. The `Lomustine` case is `experiments/lomustine_simulation`, which vendors the corrected-model table of SECQUOIA (2026b) and runs the seven methods through the harness in `experiments/catalytic_distillation_live`; the run records and analysis are in `results/lomustine_simulation_20261031`. The `Catalytic Column` model and its review are in `gdp_models/catalytic_distillation`, the live runs are `experiments/catalytic_distillation_live`, and their returned records, with the original archive, its SHA-256 checksum, and an audit of every run, are in `results/catalytic_distillation_live_20261101`. The MINLP forms, their generators, the fixed-configuration validation, and the solver runs are `experiments/catalytic_distillation_minlp`; the runs reported here are in `results/catalytic_distillation_minlp_dantzig_20261101`, and an earlier run of the same experiments on a workstation with `GAMS` 54.2.2, in `results/catalytic_distillation_minlp_20261101`, returned the same outcome for every solver, form, and configuration. The `D-SDA` neighborhood comparison of Section S14 is `experiments/dsda_separable`, with results in `results/dsda_separable_20261101`, and the check of the `Catalytic Column` configurations outside the search box is in `results/catalytic_distillation_outside_box_20261102`. The seven-method tables and figures are generated from these tables and the published records by `review_comparison/seven_methods.py` and the `review_comparison/joint_process_*.py` scripts, without new solver calls.

S9 Feasibility and bound consistency of conventional GDP results

Of the 45 conventional GDP runs in Table S23, 21 have flagged bounds (Table S14) relative to the checked solutions in Table S15. These results combine the 27 new `SCIP`–`IPOPT` GDPopt runs with the 18 retained direct `SCIP` runs. Unrounded values, residuals, and source records are linked in Section S8.

For each minimization model, let f_w be the recorded objective of its best independently checked saved feasible solution, and let L denote a reported lower bound. We apply the same rule to all five methods:

$$\tau_w = 10^{-5} \max\{1, |f_w|\}, \qquad \Delta = L - f_w, \qquad \rho = \Delta/\tau_w, \qquad \dagger \iff \rho > 1. \tag{S22}$$

Missing lower bounds remain missing, not zero.

Saved vectors are loaded into the original GDP to reevaluate objectives and check active constraints, variable bounds, integrality, disjunctions, and logical validity without optimization. Violations are unscaled original-equation residuals, not reformulation residuals. All 43 saved feasible vectors satisfy the 10^{-5} threshold; the largest violation is 9.90×10^{-7}. The two Big-M runs without accepted feasible solutions remain in the 45-run denominator. This numerical check establishes neither exact feasibility nor global optimality.

Eight of the 21 discrepancies are only 1.09–3.54 times the tolerance and are sensitive to numerical feasibility conventions. These conflicting bounds and `optimal` reports provide no independent global certificate or intrinsic method ranking; their cause in GDPopt or its subsolvers remains unresolved.

Table S14: Reported lower bounds exceeding checked feasible objectives ($\rho > 1$).

ID	Model	Method	Reported L	$v_{\max}$	C	ρ
S01	`CSTR5`	RIC	3.06201	2.60e-11	C01	1.29†
S02	`CSTR5`	GLOA	3.13020	9.85e-12	C01	2.23e+3†
S03	`CSTR5`	LOA	3.06201	2.60e-11	C01	1.29†
S05	`CSTR5`	Hull	3.06201	9.34e-7	C01	1.09†
S06	`CSTR10`	RIC	2.96605	1.62e-10	C02	2.65e+3†
S07	`CSTR10`	GLOA	3.44796	3.11e-15	C02	1.93e+4†
S08	`CSTR10`	LOA	2.96605	1.62e-10	C02	2.65e+3†
S10	`CSTR10`	Hull	2.88951	9.37e-7	C02	3.54†
S11	`CSTR15`	RIC	2.82939	1.33e-15	C03	1.30†
S12	`CSTR15`	GLOA	3.56196	1.01e-15	C03	2.59e+4†
S13	`CSTR15`	LOA	2.82939	1.33e-15	C03	1.30†
S16	`CSTR20`	RIC	2.79941	1.58e-15	C04	1.67†
S17	`CSTR20`	GLOA	3.61612	4.60e-11	C04	2.92e+4†
S18	`CSTR20`	LOA	2.79941	1.58e-15	C04	1.67†
S21	`CSTR25`	RIC	2.79125	1.38e-10	C05	3.53e+2†
S22	`CSTR25`	GLOA	3.64786	2.22e-14	C05	3.12e+4†
S27	`CSTR30`	GLOA	3.66872	7.11e-15	C06	3.25e+4†
S32	`Column`	GLOA	20355.1	1.15e-13	C07	4.65e+3†
S35	`Column`	Hull	20044.2	1.00e-8	C07	3.06e+3†
S42	`Dynamic`	GLOA	-197.140	9.85e-12	C09	7.01e+3†
S45	`Dynamic`	Hull	-196.931	1.00e-7	C09	7.11e+3†

C identifies the comparison solution in Table S15. Displayed values are rounded; flags use unrounded values.

Table S15: Checked saved comparison solutions; displayed values are rounded.

ID	Model	Method	Recorded f_w	$v_{\max}$
C01	`CSTR5`	Big-M	3.06198	9.40e-7
C02	`CSTR10`	Big-M	2.88941	9.76e-7
C03	`CSTR15`	Hull	2.82936	9.47e-7
C04	`CSTR20`	Hull	2.79936	9.90e-7
C05	`CSTR25`	Hull	2.78143	9.90e-7
C06	`CSTR30`	Big-M	2.76957	9.68e-9
C07	`Column`	RIC	19449.9	1.01e-13
C08	`Small Batch`	Big-M	167428	9.00e-7
C09	`Dynamic`	Big-M	-212.009	1.00e-7

Additional tightened-feasibility check for `CSTR15`. An `IPOPT` 3.14.19 solve retained the archived LOA witness W03's discrete configuration and starting values and targeted an original-GDP residual of 10^{-8} (SECQUOIA, 2026a). W03 is distinct from the current `SCIP` comparison solution C03. Only exactly satisfied constant rows were omitted from the computational clone; the independent check retained all original constraints. The returned objective was 2.82939, with maximum unscaled violation 1.11×10^{-16}. The new GLOA lower bound exceeds this objective by 0.732565, or approximately 2.59×10^4 times the audit tolerance. The discrepancy persists under the tighter check, but its cause remains unresolved.

Additional formulation check for `Dynamic` Hull. The saved Big-M point has objective -212.009, approximately 15.0776 below the Hull lower bound -196.931. It satisfies both the original GDP and a reconstructed Hull formulation at 10^{-5}, with maximum violation approximately 10^{-7}. The cause of the discrepancy remains unresolved.

S10 Additional candidate-search and sampling comparisons

Figure S1 shows when successful process runs first reach the target.

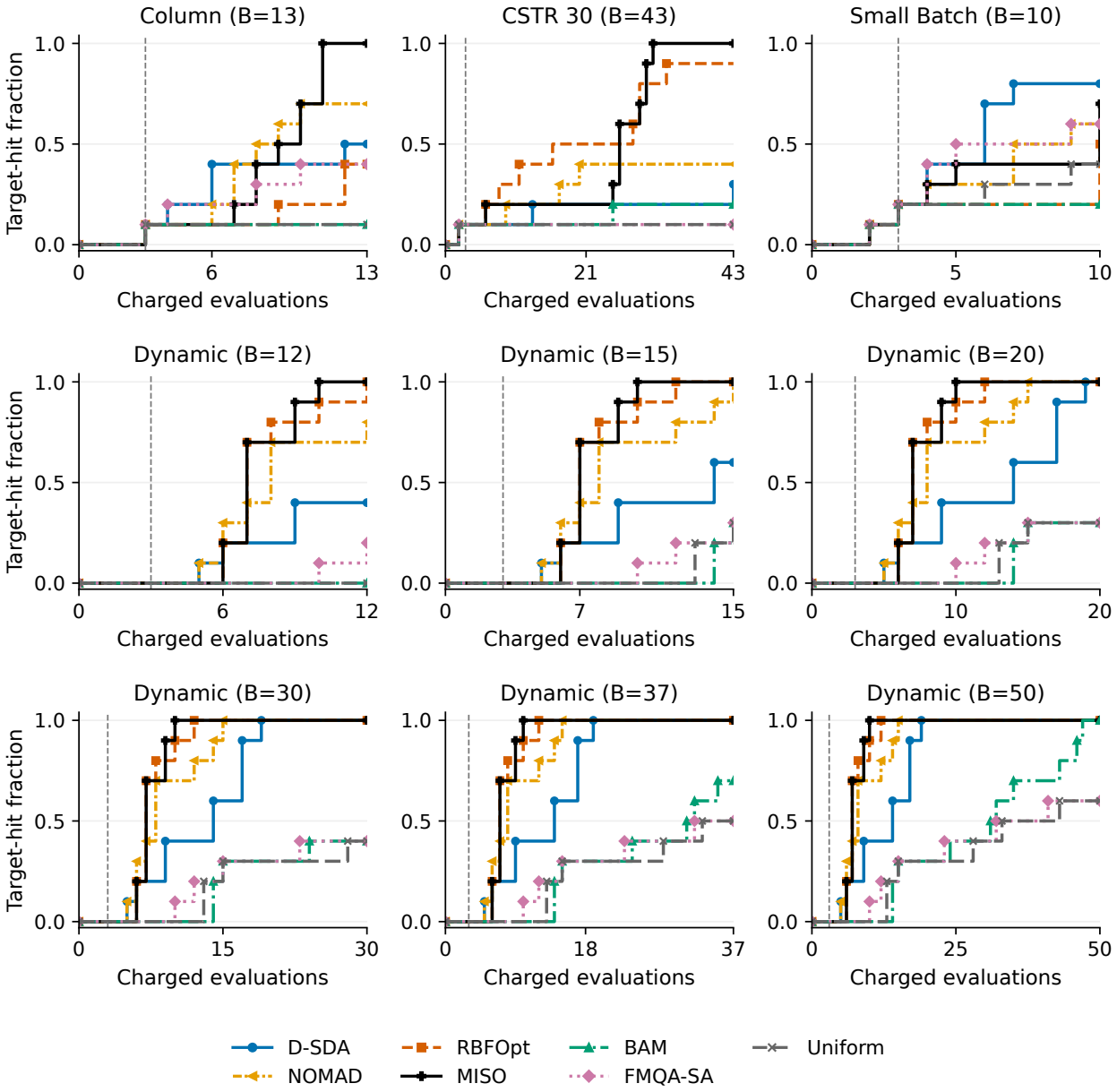


Figure S1: Cumulative target attainment versus charged evaluations for nine illustrative process conditions, using the success definition of the main paper. All ten seeds remain in every denominator, and initialization, unaccepted evaluations, and repeated configurations consume the budget. Steps mark the first accepted target observation on successful runs. The gray vertical line at three evaluations marks the end of the shared initialization. Line styles and markers distinguish methods. All 14 conditions appear in Table 2 of the main paper.

S10.1 Paired differences and sampling controls

The paired comparisons quantify differences in target-hit rates under shared initial designs, matching case, budget, and seed. Positive differences favor the first method and are expressed in percentage points. For Grid, the descriptive percentile bootstrap resamples the 21 intact function vectors, each containing ten paired seed differences, rather than treating 210 observations as exchangeable Bernoulli draws. For Process, paired seeds are resampled within each of the 14 conditions; the six related `Dynamic` budgets are not pooled for inference. The seed is 20261020 with 2,000 bootstrap replications and deterministic case/seed ordering. Intervals are exploratory and unadjusted for multiple comparisons; ten-seed boundary cases may yield degenerate intervals, and zero inclusion does not establish equivalence.

An interval entirely above zero favors the first method under this resampling; an interval spanning zero leaves the direction uncertain. There is no preset cutoff for practical importance. For example, `BAM` matches its prefix control at $B = 10$ but exceeds it by 10.95 percentage points at $B = 100$, so initial sampling alone does not account for its final target-hit rate.

Table S16: Selected paired success differences, in percentage points. The first method minus the second defines the sign; Process intervals concern the stated condition only.

Grid

B	Contrast	Difference	95% interval
10	`RBFOpt` – Uniform	13.33	[5.71, 24.29]
100	`RBFOpt` – Uniform	60.00	[42.86, 75.71]
10	`BAM` – `BAM`-prefix control	0.00	[0.00, 0.00]
100	`BAM` – `BAM`-prefix control	10.95	[0.48, 23.81]
10	`FMQA-SA` – Uniform	0.95	[-0.95, 2.86]
100	`FMQA-SA` – Uniform	40.00	[28.57, 52.38]
10	`D-SDA` – `RBFOpt`	-13.81	[-24.29, -6.19]
100	`D-SDA` – `RBFOpt`	-33.33	[-48.10, -19.52]

Process: ten paired seeds per condition

Case	B	Contrast	Outcomes	Difference	95% interval
`Dynamic`	12	`RBFOpt` – `D-SDA`	4/6/0/0	60	[30, 90]
`Small Batch`	10	`FMQA-SA` – Uniform	2/4/2/2	20	[-30, 60]

Outcomes list both successful / first only / second only / neither successful.

S10.2 Exact paired tests for process conditions

Table S17 lists the seed-paired target outcomes of `RBFOpt` and `D-SDA` in every process condition and of `BAM` and `RBFOpt` on `CSTR5`. The p-values come from the two-sided exact McNemar test, which uses only the pairs in which exactly one method reaches the target. They are descriptive and not adjusted for multiple comparisons. In the 60 paired `CSTR` runs, `RBFOpt` reaches the target in 30 of the 31 discordant pairs.

Table S17: Seed-paired target outcomes and exact McNemar p-values. First only and second only count seeds in which only the first or the second method reaches the target.

Contrast	Case	B	Both	First only	Second only	Neither	p
RBFOpt vs D-SDA	`CSTR5`	8	2	6	0	2	0.0312
RBFOpt vs D-SDA	`CSTR10`	15	2	4	1	3	0.375
RBFOpt vs D-SDA	`CSTR15`	22	2	5	0	3	0.0625
RBFOpt vs D-SDA	`CSTR20`	29	3	4	0	3	0.125
RBFOpt vs D-SDA	`CSTR25`	36	3	5	0	2	0.0625
RBFOpt vs D-SDA	`CSTR30`	43	3	6	0	1	0.0312
RBFOpt vs D-SDA	`CSTR`, all sizes	–	15	30	1	14	2.98×10^{-8}
RBFOpt vs D-SDA	`Column`	13	3	1	2	4	1
RBFOpt vs D-SDA	`Small Batch`	10	4	1	4	1	0.375
RBFOpt vs D-SDA	`Dynamic`	12	4	6	0	0	0.0312
RBFOpt vs D-SDA	`Dynamic`	15	6	4	0	0	0.125
RBFOpt vs D-SDA	`Dynamic`	20	10	0	0	0	1
RBFOpt vs D-SDA	`Dynamic`	30	10	0	0	0	1
RBFOpt vs D-SDA	`Dynamic`	37	10	0	0	0	1
RBFOpt vs D-SDA	`Dynamic`	50	10	0	0	0	1
BAM vs RBFOpt	`CSTR5`	8	8	2	0	0	0.500

S10.3 Grid suite exclusions

Excluding Xin-She Yang 3 and Bohachevsky 2 preserves the main descriptive pattern. For the sampled Bohachevsky 2 lattice, $\cos(3\pi a_i) = 1$ at every coordinate, leaving an objective nearly redundant with sampled Sum of Squares after normalization. Without both functions, $B = 100$ rates are 84.7% for `MISO`, 79.5% for `RBFOpt`, 66.8% for `BAM`, 57.9% for `FMQA-SA`, 55.8% for `NOMAD`, 54.7% for the `BAM`-prefix control, 42.6% for `D-SDA`, and 17.9% for uniform. At $B = 10$, `BAM` and its prefix control each retain 47.9%, compared with 15.8% for `RBFOpt`, 14.7% for `MISO`, and 12.6% for `NOMAD`.

S11 Seed spread, tolerance sensitivity, and search overhead

Table S18 compares exact target attainment with relaxed relative tolerances for the 980 primary process runs. The relative gap is $(f_{\text{best}} - f_{\text{ref}})/\max\{1, |f_{\text{ref}}|\}$, the normalization of the target rule. The exact column applies the target tolerance of 10^{-5}. Distinct finals is the largest number of different final objectives among the ten seeds of any condition in the family, and the miss-gap range covers every run that misses the exact target. Table S19 compares the median configuration-evaluation cost with the median search time per evaluation. Both are medians over runs of per-run averages, namely the summed point cost or the stored search time divided by the number of evaluations, over all seven methods for the evaluation cost. The 21 `NOMAD` Grid runs without a recorded search time are left out of its search median.

Every NLP subproblem solved by `GDPopt` in the process comparisons fixes a different configuration, as recorded in its execution logs before each subproblem. RIC and LOA solve one subproblem for each of the $N(N+1)/2$ logically admissible configurations of `CSTR5` to `CSTR20` and for each of the 37 admissible `Dynamic` configurations, and RIC also covers all 325 admissible configurations of `CSTR25`.

Table S18: Exact and relaxed target attainment and spread of final objectives across seeds. The relaxed columns count runs whose relative gap to f_{ref} is at most the stated value.

Family	Method	Exact	$\leq 0.1\%$	$\leq 1\%$	$\leq 5\%$	Distinct finals	Miss gap range (%)
	D-SDA	16/60	16	19	27	5	0.224–13.0
	NOMAD	21/60	21	21	27	3	2.23–13.0
	RBFOpt	45/60	45	45	47	3	2.23–12.5
`CSTR`	MISO	43/60	43	43	49	5	2.23–10.6
	BAM	18/60	18	60	60	3	0.224–0.692
	FMQA-SA	9/60	9	23	41	10	0.169–17.2
	Uniform	11/60	11	25	48	10	0.224–14.3
	D-SDA	5/10	5	8	10	4	0.536–1.14
	NOMAD	7/10	7	9	10	3	0.536–1.14
	RBFOpt	4/10	4	5	9	6	0.536–5.92
`Column`	MISO	10/10	10	10	10	1	–
	BAM	1/10	1	1	10	2	1.05–1.05
	FMQA-SA	4/10	4	4	10	4	1.05–2.76
	Uniform	1/10	1	2	9	8	0.536–5.22
	D-SDA	8/10	8	8	8	2	22.2–22.2
	NOMAD	6/10	6	6	6	2	6.64–6.64
	RBFOpt	5/10	5	5	5	3	6.64–22.2
`Small Batch`	MISO	7/10	7	7	7	2	6.64–6.64
	BAM	2/10	2	2	2	3	6.64–8.23
	FMQA-SA	6/10	6	6	6	5	6.64–22.2
	Uniform	4/10	4	4	4	4	6.64–22.2
	D-SDA	50/60	50	50	54	5	2.07–36.3
	NOMAD	58/60	58	58	60	3	2.07–2.66
	RBFOpt	60/60	60	60	60	1	–
`Dynamic`	MISO	60/60	60	60	60	1	–
	BAM	27/60	27	27	46	4	2.07–7.01
	FMQA-SA	23/60	23	23	42	7	2.07–36.3
	Uniform	21/60	21	21	38	6	2.07–30.7

Table S19: Median configuration-evaluation cost and median search time per evaluation.

		Search time per evaluation (ms)						
Family	Evaluation cost (s)	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
Grid-GDP	0.390	0.0777	50.5	31.1	231	3.77	604	0.0336
`CSTR`	1.46	1.24	72.2	24.2	287	9.01	572	0.517
`Column`	2.50	0.941	3.25	17.4	51.7	1.36	527	0.551
`Small Batch`	0.422	1.02	4.75	17.0	66.8	1.69	487	0.536
`Dynamic`	0.836	0.893	96.9	98.4	1860	9.34	567	0.471

S12 NOMAD and MISO runs

The NOMAD and MISO runs use the stored responses, shared initial configurations, seeds, budgets, penalties, and reference targets of the other five methods. Neither method receives the target as a stopping criterion. They were executed after the other methods, on a shared server with six workers, each restricted to one logical CPU on a different physical core and to one OpenMP and BLAS thread; the cores were not reserved exclusively. The server used Python 3.11.16, PyNomadBBO 4.6.0, soogo 2.1.0, NumPy 2.4.6, and SciPy 1.17.1. As for the other methods, completion times add the measured search time of each run to the recorded cost of every evaluated configuration, so they do not depend on the speed of the server's NLP solves.

Termination. All 1,190 MISO runs use their full budget. NOMAD stops before the budget in 77 of the 140 process runs and in 697 of the 1,050 Grid runs, including all 210 runs at $B = 100$, where it stops after a median of 31.5 evaluations. Its Grid target fraction is therefore 60.0% at both $B = 70$ and $B = 100$.

NOMAD **crashes and reruns.** Of the 1,190 NOMAD runs, 1,046 completed on the first attempt. The other 144 ended with a segmentation fault after their last new evaluation. Of these, 110 completed on the first rerun and 13 on the second, each after evaluating the same configurations in the same order as the crashed attempt, and 21 crashed on all three attempts after identical evaluations. All 21 are Grid runs at $B = 100$, and 16 of them had reached the target before the crash. Scoring them on the evaluations completed before the crash gives NOMAD 526 of 1,050 Grid targets and 92 of 140 process targets; counting all 21 as misses gives 510 Grid targets, or 52.4% instead of 60.0% at $B = 100$. These 21 runs have no recorded search time, so the NOMAD timing summaries at Grid $B = 100$ use the other 189 runs. For a completed rerun, the search time is that of the completed attempt; the time spent in crashed attempts is not recorded. No MISO run failed.

Exploratory paired tests. Table S20 compares NOMAD and MISO with D-SDA, RBFOpt, and each other in every process condition, a family of 70 exact McNemar tests defined after the runs were completed. With a Holm correction over the 70 tests, the smallest adjusted p-value is 0.547, for MISO against NOMAD on CSTR20, so no contrast is significant at the 0.05 level. Ten seeds give each test little power, and a nonsignificant test does not establish equivalence. The six Dynamic budgets reuse the same seeds and are not independent replications. The last row pools the six CSTR sizes for the descriptive comparisons in the main text; it is not part of the corrected family.

Table S20: Seed-paired target outcomes of NOMAD and MISO. Each cell gives the number of seeds in which only the first or only the second method reaches the target, and the unadjusted two-sided exact McNemar p-value.

Case	B	NOMAD vs D-SDA	NOMAD vs RBFOpt	MISO vs D-SDA	MISO vs RBFOpt	MISO vs NOMAD
Column	13	3:1 0.625	4:1 0.375	5:0 0.0625	6:0 0.0312	3:0 0.250
CSTR5	8	2:0 0.500	1:5 0.219	2:0 0.500	0:4 0.125	1:1 1
CSTR10	15	0:0 1	1:4 0.375	0:1 1	0:4 0.125	0:1 1
CSTR15	22	0:0 1	0:5 0.0625	5:0 0.0625	1:1 1	5:0 0.0625
CSTR20	29	0:1 1	0:5 0.0625	7:0 0.0156	3:0 0.250	8:0 0.00781
CSTR25	36	4:1 0.375	2:4 0.688	7:0 0.0156	2:0 0.500	4:0 0.125
CSTR30	43	1:0 1	1:6 0.125	7:0 0.0156	1:0 1	6:0 0.0312
Small Batch	10	1:3 0.625	2:1 1	1:2 1	2:0 0.500	3:2 1
Dynamic	12	4:0 0.125	0:2 0.500	6:0 0.0312	0:0 1	2:0 0.500
Dynamic	15	4:0 0.125	0:0 1	4:0 0.125	0:0 1	0:0 1
Dynamic	20	0:0 1	0:0 1	0:0 1	0:0 1	0:0 1
Dynamic	30	0:0 1	0:0 1	0:0 1	0:0 1	0:0 1
Dynamic	37	0:0 1	0:0 1	0:0 1	0:0 1	0:0 1
Dynamic	50	0:0 1	0:0 1	0:0 1	0:0 1	0:0 1
CSTR, all sizes	–	7:2 0.180	5:29 3.86×10^{-5}	28:1 1.12×10^{-7}	7:9 0.804	24:2 1.05×10^{-5}

S13 Process D-SDA restarts

The process `D-SDA` of the main comparison descends once from the best of the three shared initial configurations (S). Variant R applies the Grid rule and descends from all three shared initial configurations, best first, while budget remains. Variant RR continues after R with descents from uniformly drawn unvisited configurations until the budget is exhausted, and it uses the full budget in all 140 runs. No variant receives the reference target. All variants read the stored responses and, like the main comparison, never evaluate a configuration twice. Table S21 shows that R and RR evaluate 1,055 and 1,249 more configurations than S and add one target, on seed 4 of `CSTR30`, so target attainment rises from 79 to 80 of the 140 runs. Both restart variants improve the final objective in the same 14 `CSTR` runs. In each of them, S ends at $(5, 1)$, 8.3 to 13.0% above the reference, and the restarts end at the target in the one new hit and 0.4 to 8.6% above it in the other 13. On `CSTR10` to `CSTR30`, 31 of the 36 single-descent misses end at $(5, 1)$.

Table S21: Process `D-SDA` with a single descent (S), restarts from the ranked shared initial configurations (R), and additional random restarts (RR). Completion times add the measured search time to the recorded configuration costs.

		Target hits			Full-budget runs		Median evaluations			Median completion (s)		
Case	B	S	R	RR	S	R	S	R	RR	S	R	RR
`Column`	13	5	5	5	9	10	13	13	13	35.44	35.69	35.69
`CSTR5`	8	2	2	2	8	10	8	8	8	10.67	10.71	10.71
`CSTR10`	15	3	3	3	3	10	11.5	15	15	17.61	21.57	21.57
`CSTR15`	22	2	2	2	2	10	15.5	22	22	25.68	35.19	35.19
`CSTR20`	29	3	3	3	0	10	19	29	29	32.31	47.49	47.49
`CSTR25`	36	3	3	3	2	9	23.5	36	36	45.49	60.78	60.78
`CSTR30`	43	3	4	4	1	8	24.5	43	43	47.17	78.88	78.88
`Small Batch`	10	8	8	8	10	10	10	10	10	4.23	4.23	4.23
`Dynamic`	12	4	4	4	9	10	12	12	12	12.22	12.25	12.25
`Dynamic`	15	6	6	6	6	10	15	15	15	15.32	15.76	15.76
`Dynamic`	20	10	10	10	3	10	18	20	20	20.02	20.76	20.76
`Dynamic`	30	10	10	10	0	10	18	30	30	20.02	27.13	27.13
`Dynamic`	37	10	10	10	0	2	18	34.5	37	20.02	28.01	30.05
`Dynamic`	50	10	10	10	0	0	18	34.5	50	20.02	28.01	37.00

S14 D-SDA neighborhood

The process, Grid, and `Lomustine D-SDA` runs use the unit L_∞ neighborhood, whose $3^d - 1$ neighbors number 8 for $d = 2$ external variables and 26 for $d = 3$. The `Catalytic Column`, with $d = 6$ and 728 such neighbors, uses the separable neighborhood, which changes one external variable by ± 1 and has at most $2d$ neighbors. To test the effect of this choice, we repeated every `D-SDA` run on stored responses with the separable neighborhood and all other settings unchanged: the 1,050 Grid runs, the 140 single-descent process runs, and the 180 `Lomustine` runs in the six option orders of Section S5. Table S22 pairs each run with its L_∞ counterpart.

The separable neighborhood reaches more targets when the budget is small compared with the L_∞ neighborhood: on Grid at $B = 10$ (22 against 13 of 210 runs, with 9 and 0 discordant runs, exact McNemar $p = 0.004$), on `Dynamic` at $B = 12$ and 15 (16 against 10 of 20, $p = 0.031$), and on `Lomustine` over the six orders (99 against 90 of 180, $p = 0.022$). Its descents also stop earlier: from $B = 50$, the Grid runs end after a median of 25 evaluations in place of 37, and they reach 88 targets against 101 at $B = 70$ and 100 ($p = 0.011$). On `CSTR`, the separable neighborhood reaches 7 of the 60 targets against 16 ($p = 0.004$). Eighteen of its 53 misses stop on the edge $\xi_1 = \xi_2$ of the admissible region, where every one-coordinate move leaves the region or worsens the objective, whereas a diagonal move continues along the edge to the target at (N, N). Over the 140 process runs, the two neighborhoods reach 79 and 78 targets. Neither neighborhood dominates: the L_∞ neighborhood suits two external variables with budgets that cover its eight neighbors, and the separable neighborhood suits more external variables or smaller budgets.

Table S22: Target hits of `D-SDA` with the unit L_∞ and the separable neighborhoods, the runs that only one of them solves, and the median number of evaluations. Each process row has ten paired runs. Bold counts mark the larger number of hits.

Case	B	L_∞	Separable	Separable only	L_∞ only	Median evals L_∞	Median evals separable
Grid (210 runs)	10	13	**22**	9	0	10	10
Grid (210 runs)	30	78	**79**	14	13	30	25
Grid (210 runs)	50	**91**	87	10	14	37	25
Grid (210 runs)	70	**101**	88	5	18	37	25
Grid (210 runs)	100	**101**	88	5	18	37	25
`Column`	13	5	**6**	3	2	13	12
`CSTR5`	8	2	2	0	0	8	8
`CSTR10`	15	**3**	1	0	2	11.5	8
`CSTR15`	22	**2**	1	0	1	15.5	10.5
`CSTR20`	29	**3**	1	0	2	19	14
`CSTR25`	36	**3**	1	0	2	23.5	15.5
`CSTR30`	43	**3**	1	0	2	24.5	17
`Small Batch`	10	8	**9**	2	1	10	10
`Dynamic`	12	4	**6**	2	0	12	12
`Dynamic`	15	6	**10**	4	0	15	13
`Dynamic`	20	10	10	0	0	18	13
`Dynamic`	30	10	10	0	0	18	13
`Dynamic`	37	10	10	0	0	18	13
`Dynamic`	50	10	10	0	0	18	13
`Lomustine`, six orders (60 runs)	10	23	**27**	6	2	10	10
`Lomustine`, six orders (60 runs)	15	32	**36**	4	0	15	11
`Lomustine`, six orders (60 runs)	20	35	**36**	1	0	19	11

S15 Joint comparison with conventional GDP methods

Figure S2 compares the eight process conditions not shown in the main joint completion-time figure. Each DFO condition has ten seeds, whereas each conventional GDP method has one run per problem. Timing and target criteria follow Figure 2 of the main paper. Table S23 reports all nine models, including final objectives and termination flags.

Table S23: Full-process-GDP results from one run per method and model. Entries report the final objective and measured full-run completion time (s) in parentheses, including model-template preparation; runs stopped by the 1,800 s time limit are shown as 1800+. `GDPopt` uses `SCIP` for master problems and `IPOPT` for NLP subproblems; Big-M and Hull retain their `SCIP` results. Bold marks the lowest objective in each row and ties within $10^{-5}\max\{1, |f_{\min}|\}$, where $f_{\min}$ is the lowest objective in that row. Feasibility means numerical feasibility under the original-GDP residual test at 10^{-5}, not exact feasibility.

Model	RIC	GLOA	LOA	Big-M	Hull
`CSTR5`	3.06201† (15.38)	3.13020†‡ (4.57)	3.06201† (24.58)	**3.06198** (53.68)	3.06201† (15.43)
`CSTR10`	2.96605†‡ (134.31)	3.44796†‡ (3.10)	2.96605†‡ (140.38)	**2.88941** (771.36)	2.88951† (71.63)
`CSTR15`	2.82939† (283.77)	3.56196†‡ (6.17)	2.82939† (445.43)	-^{TL} (1800+)	**2.82936** (284.63)
`CSTR20`	2.79941† (607.72)	3.61612†‡ (7.46)	2.79941† (733.53)	2.80413^{TL} (1800+)	**2.79936** (1150.67)
`CSTR25`	2.79125†‡ (1042.79)	3.64786†‡ (5.20)	2.79125^{TL} (1800+)	2.78771^{TL} (1800+)	**2.78143** (1701.75)
`CSTR30`	2.77162^{TL} (1800+)	3.66872†‡ (5.52)	2.87314^{TL} (1800+)	**2.76957**^{TL} (1800+)	**2.76957**^{TL} (1800+)
`Column`	**19449.9** (124.03)	20355.1†‡ (42.41)	**19449.9** (199.22)	-^{TL} (1800+)	22271.5$^{\mathrm{TL}\dagger}$ (1800+)
`Dynamic`	**-212.009** (149.39)	-197.140†‡ (238.96)	**-212.009** (550.49)	**-212.009**^{TL} (1800+)	-196.931†‡ (921.28)
`Small Batch`	**167428** (13.15)	**167428** (10.32)	**167428** (5.61)	**167428** (0.51)	**167428** (0.50)

TL: reported `maxTimeLimit` under the configured 1,800 s limit. Runs without TL report `optimal`.
Dash: no accepted feasible solution.
†: reported lower bound exceeds the best checked saved numerically feasible objective f by more than $10^{-5}\max\{1, |f|\}$.
‡: reference target missed despite reported `optimal` and a better checked numerically feasible solution.

Table S24: NLP workload in the process comparisons with `SCIP`–`IPOPT` GDPopt. DFO entries are median counts of corresponding fixed-configuration NLP evaluations over ten seeds; GDPopt entries count confirmed IPOPT NLP interface calls in one run. Logical and preprocessing rejections are excluded; rejected NLP returns are retained. These counts are not internal solver iterations or equally costly subproblems.

Case	B	D-SDA	RBFOpt	BAM	FMQA-SA	Uniform	GLOA	LOA	RIC
CSTR5	8	6	6	6	5.5	4.5	4	15	15
CSTR10	15	10	12	8.5	9	7.5	2	55	55
CSTR15	22	17	16.5	11	15	11	4	120	120
CSTR20	29	21	20.5	17	17.5	15.5	4	210	210
CSTR25	36	25	25.5	18.5	23	21	2	315	325
CSTR30	43	26.5	31.5	21.5	30.5	21.5	2	243	455
Column	13	11	12	7.5	7	7	18	38	38
Small Batch	10	4.5	6	5	5	4	12	5	12
Dynamic	12	10	10	8	6.5	6	20	37	37
Dynamic	15	13	13	9	8	6.5	20	37	37
Dynamic	20	17.5	16	11	11	9.5	20	37	37
Dynamic	30	22	21	17	14.5	15	20	37	37
Dynamic	37	22	25	19	19	19	20	37	37
Dynamic	50	22	31	26	24.5	26	20	37	37

The same single Dynamic GDPopt execution is repeated across budget rows for reference. Internal NLP counts for direct Big-M and Hull are unavailable and are not treated as zero. Every `GDPopt` NLP call fixes a different configuration; `CSTR`N has $N(N+1)/2$ logically admissible configurations, and `Dynamic` has 37 (Section S11).

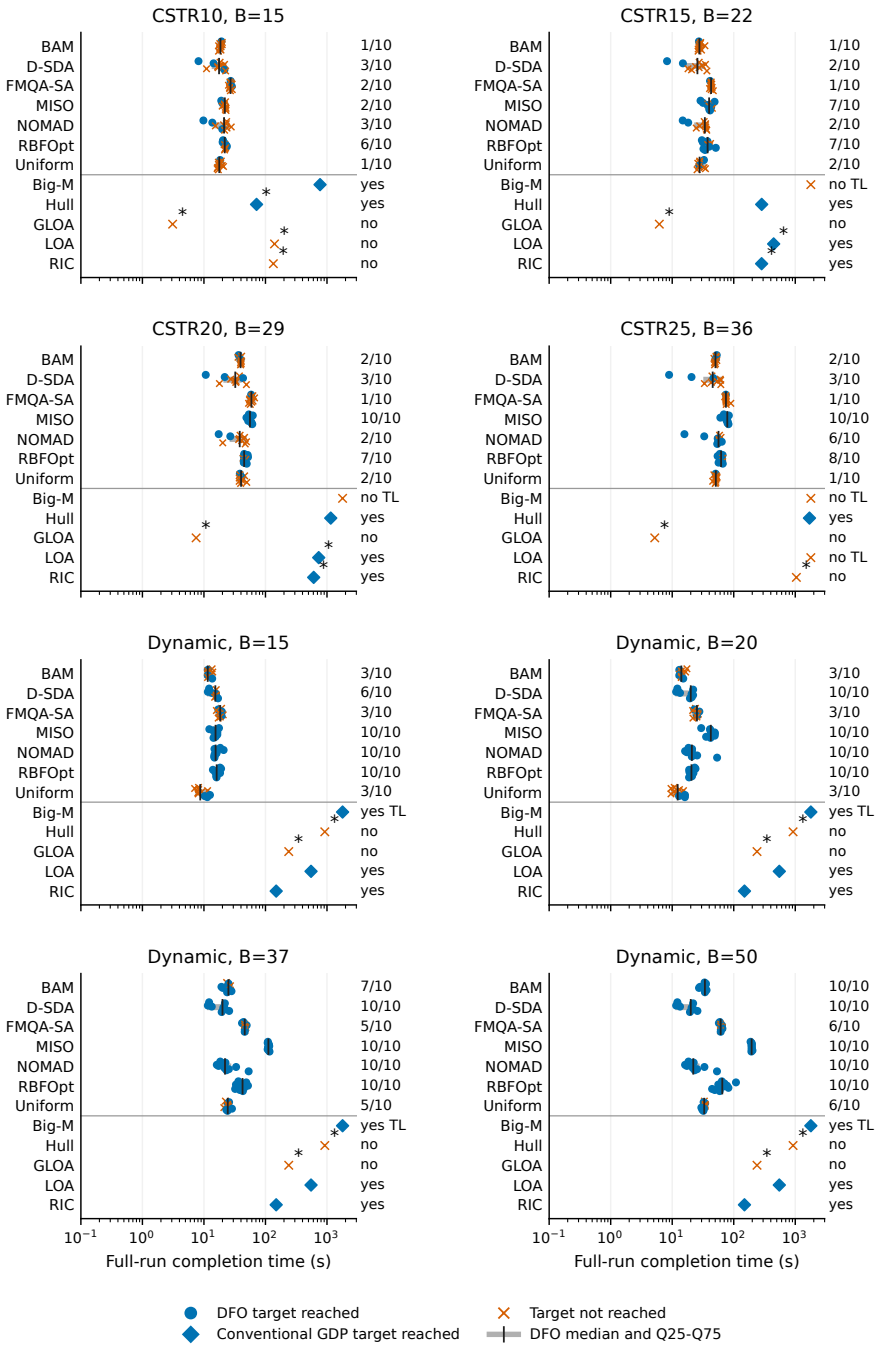


Figure S2: Joint full-run completion-time comparison for the eight additional process conditions. Dots show successful DFO seeds, diamonds successful single conventional runs, and crosses target misses. Gray segments and ticks show Q25–Q75 and medians across all ten DFO seeds. TL denotes time-limit termination; asterisks mark bound inconsistencies (Section S9). The Dynamic panels reuse the same conventional observations.

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